MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, AMPLIFIER
TYPE 2N4957
JAN, JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, RF transistors. Four levels of product assurance are provided for the device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	V_{CEO}	V_{CBO}	V_{EBO}	I_C	T_{STG} and T_J
mW	V dc	V dc	V dc	mA dc	°C
200	30	30	3.0	30	-65 to +200

1/ Derate at 1.14 mW/°C above 25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE3}	h_{fe}	$r_b'C_c$	C_{cb}	G_{pe}	F
	I_C = 5.0 mA dc V_{CE} = 10 V dc	I_E = 2.0 mA dc V_{CE} = 10 V dc f = 100 MHz	I_E = 2.0 mA dc V_{CB} = 10 V dc f = 63.6 MHz	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤1 MHz	I_C = 2.0 mA dc V_{CE} = 10 V dc f = 450 MHz	I_C = 2.0 mA dc V_{CE} = 10 V dc f = 450 MHz
	---	---	ps	pF	dB	dB
Min	30	12	1.0	---	17	---
Max	165	36	8.0	0.8	25	3.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rome Air Development Center, ATTN: RBE-2, Griffis AFB, NY 13441 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

C_{cb} - - - - - Collector-to-base feedback capacitance

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold-plated or tinned. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the transistor:

a. Country of origin.

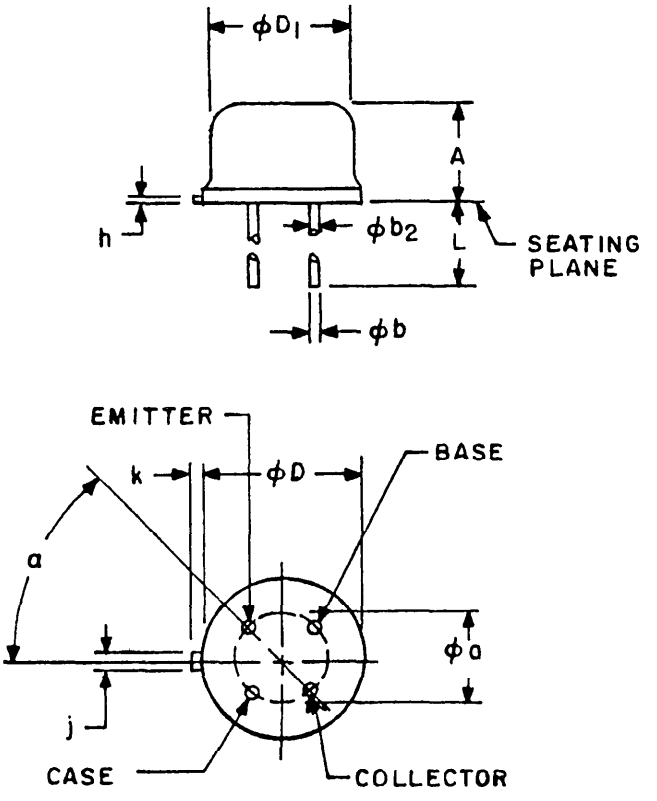
b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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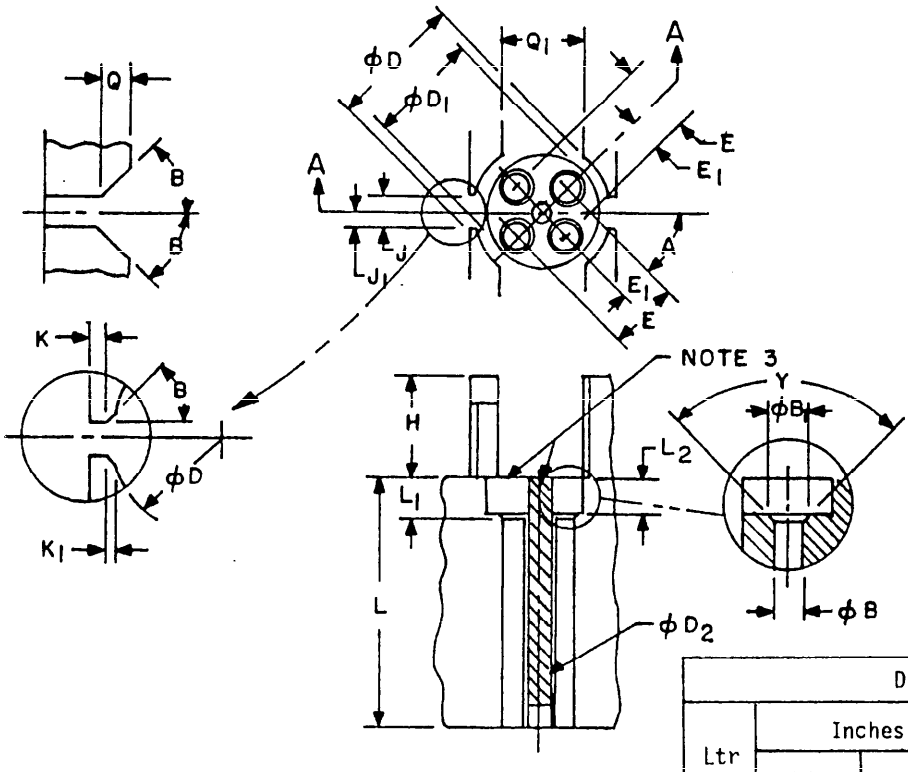


Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.170	.210	4.32	5.33	
Øa	.100 Nom		2.54 Nom		4
Øb	.016	.021	0.41	0.53	2,7
$Øb_2$	.016	.019	0.41	0.48	3,7
ØD	.209	.230	5.31	5.84	
$ØD_1$	.178	.195	4.52	4.95	
h		.020		0.51	
j	.036	.046	0.91	1.17	
k	.028	.048	0.71	1.22	6
L	.500	.750	12.70	19.05	7
α		45° Nom			

NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, -.000 (1.37 +.03, -.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 is the preferred measuring method.
5. Emitter, Base, and Collector leads electrically insulated from the case. The case lead is electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All 4 leads.

FIGURE 1. Physical dimensions.

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SECTION A-A

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	.0325	.0335	0.826	0.851	4
$ØB_1$	.043 Nom		1.09 Nom		4
ØD	.2310	.2315	5.867	5.880	
$ØD_1$	.159	.161	4.04	4.09	
$ØD_2$	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.527	2.553	
E_1	.0495	.0505	1.257	1.283	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.194	1.206	
J_1	.0235	.0245	0.597	0.622	
K	.009	.011	0.229	0.279	
K_1	.005 Nom		0.13 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90° Min		45.10° Max		
B		45° Nom			
Y		90° Nom			

NOTES:
1. Metric equivalents are given for general information only.
2. The following gauging procedure shall be used: The device being measured shall be inserted until its seating plane is .125 (3.18 mm) + .010 (0.25 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE3}	Not applicable
11	I_{CBO1} and h_{FE3}; ΔI_{CBO1} = 100% or 10 nA dc, whichever is greater; Δh_{FE3} = 15%	I_{CBO1} and h_{FE3}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% or 10 nA dc, whichever is greater; Δh_{FE3} = 15%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% or 10 nA dc, whichever is greater; Δh_{FE3} = 15%

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

P_T = 200 mW, V_{CB} = 24 V dc

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Collector-base time constant. This parameter may be determined by applying an RF signal voltage of 1 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{eb}) with a high impedance rf voltmeter across the emitter-base terminals. With f = 63.6 MHz used for the 1 volt signal, the following computation applies: $r_b'C_c$ (ps) = 2 x V_{eb} (millivolts).

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN JANTX JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Collector-to-emitter breakdown voltage	3011	Bias condition D; I_C = 1.0 mA dc; I_B = 0			$V_{(BR)CEO}$	30		V dc
Collector-to-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; I_E = 0			$V_{(BR)CBO}$	30		V dc
Emitter-to-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; I_C = 0			$V_{(BR)EBO}$	3.0		V dc
Collector-to-base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc; I_E = 0			I_{CBO1}		100	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.5 mA dc			h_{FE1}	15		
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 2.0 mA dc			h_{FE2}	20		
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5.0 mA dc			h_{FE3}	30	165	
Subgroup 3								
High-temperature operation:		T_A = 150°C						
Collector-to-base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc; I_E = 0			I_{CBO2}		100	µA dc
Low-temperature operation:		T_A = -55°C						
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5 mA dc			h_{FE4}	10		
Subgroup 4								
Magnitude of common emitter small signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_E = 2.0 mA dc; f = 100 MHz; case lead grounded			$\lvert h_{fe} \rvert$	12	36	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JAN JANTX JANTXV		Min	Max	
Collector-to-base feedback capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤1 MHz; case and emitter leads shall be guarded			C_{cb}		0.8	pF
Collector-to-base time constant		V_{CB} = 10 V dc; I_E = 2.0 mA dc; f = 63.6 MHz; case lead shall be grounded (see 4.5.1 and figure 3)			$r_b'C_c$	1.0	8.0	ps
Noise figure	3246	V_{CE} = 10 V dc; I_C = 2.0 mA dc; f = 450 MHz; R_L = 50 ohms; case lead shall be grounded (see figure 4)			F		3.0	dB
Common-emitter small signal power gain	3256	V_{CE} = 10 V dc; I_C = 2.0 mA dc; f = 450 MHz; case lead shall be grounded (see figure 4)			G_{pe}	17	25	dB
Subgroups 5, 6 and 7								
Not applicable								

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
Fine leak				
Gross leak				
Electrical measurements		See table IV, steps 1 and 4		
Decap internal visual (design verification)	2075			
Bond strength	2037	Test condition A		
Die shear strength	2017			
Subgroup 4				
Intermittent operation life	1037	V_{CB} = 20 V dc; P_T = 200 mW at T_A = 25°C ±3°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2000 cycles. No heat sink or forced air cooling on the devices shall be permitted		
Electrical measurements		See table IV, steps 3 and 5		
Subgroup 5				
Accelerated steady state operation life	1027	V_{CB} = 20 V dc; T_A = 125°C ±25°C for 96 hours; P_T = 200 mW at 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply		

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Electrical measurements		See table IV, steps 3 and 5		
Bond strength (Al-Au Die interconnects only).	2037	Test condition A		
Subgroup 6				
Not applicable				

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
Fine leak			
Gross leak			
Electrical measurements		See table IV, steps 1 and 4	
Subgroup 3			
Steady-state operation life	1027	V_{CB} = 20 V dc; P_T = 200 mW No heat sink or forced air cooling on the devices shall be permitted.	
Electrical measurements		See table IV, steps 2 and 4	
Subgroup 4			
Decap internal visual (design verification)	2075		
Bond strength	2037	Test condition A	
Subgroup 5			
Not applicable			
Subgroup 6			
High-temperature life (non-operating)	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 2 and 4	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE III. Group C inspection (all quality levels).

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 1			
Physical dimensions	2066	See figure 1	
Subgroup 2			
Thermal shock (glass strain)	1056		
Terminal strength	2036	Test condition E	
Hermetic seal	1071		
Fine leak			
Gross leak			
Moisture resistance	1021		
External visual	2071		
Electrical measurements		See table IV, steps 1 and 4	
Subgroup 3			
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1 and 4	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			
Steady-state operation life	1026	T_A = 25°C; V_{CB} = 20 V dc; P_T = 200 mW	
Electrical measurements		See table IV, steps 2 and 4	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Groups A, B, C, electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector-to-base cutoff current	3036	Bias Condition D; V_{CB} = 20 V dc; I_E = 0	I_{CBO1}		100	nA dc
2.	Collector-to-base cutoff current	3036	Bias Condition D; V_{CB} = 20 V dc; I_E = 0	I_{CBO1}		150	nA dc
3.	Collector-to-base cutoff current	3036	Bias Condition D; V_{CB} = 20 V dc; I_E = 0	ΔI_{CBO1}		±100% of initial value or 10 nA dc, whichever is greater	
4.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5 mA dc	h_{FE3}	30	165	
5.	Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5 mA dc	Δh_{FE3}		±20% change from initial reading	

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TEKTRONIX MODEL 190A
SIGNAL GENERATOR
(OR EQUIVALENT)

BOONTON
UNTERMINATED PROBE
TYPE 91-6C
AND RF VOLT-METER
MODEL 91CA
(OR EQUIVALENT)

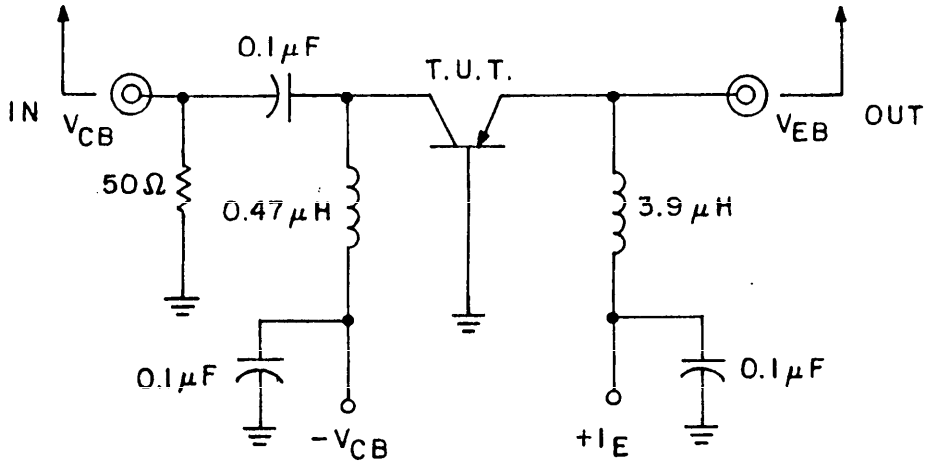
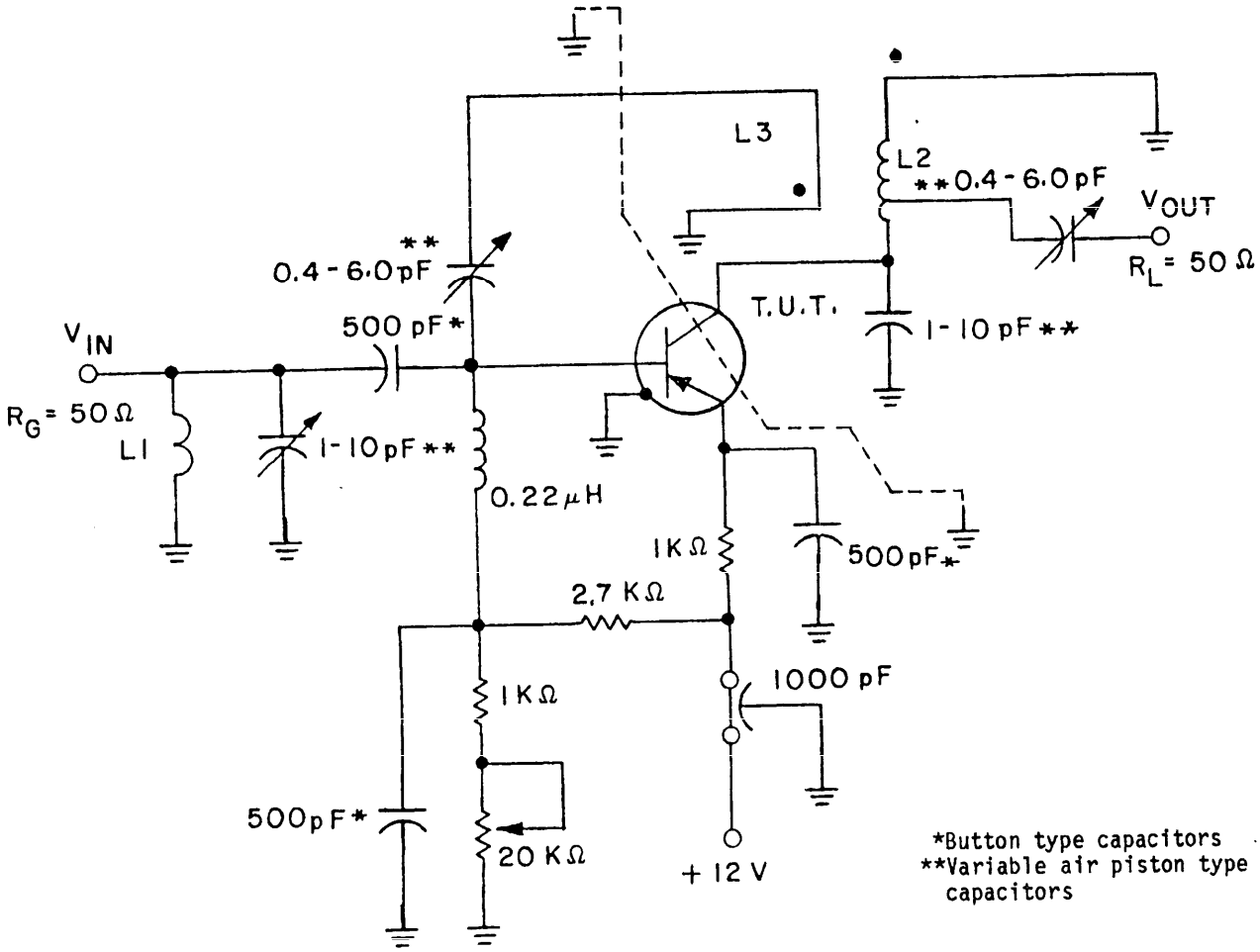


FIGURE 3. $r_b'C_c$ circuit.

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NOTES:
1. L1 - Silver-plated brass bar, 1.0 inch long by 0.25 inch od. (Straight bar.)
2. L2 - Silver-plated brass bar, 1.5 inches long by 0.25 inch od. Tap is 0.25 inch from collector. (Straight bar.)
3. L3 - 1/2 turn of AWG number 16 wire, loop od. approximately 0.5 inch, located 0.25 inch from, and parallel to L2.
4. The noise source is a hot-cold body, (AIL type 70 or equivalent) with a test receiver (AIL type 136 or equivalent).

FIGURE 4. F and G_{pe} test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents may specify the lead material and finish (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
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Air Force - 11, 19, 80, 85
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/426A	Semiconductor Device, Transistor, PNP, Silicon, Amplifier Type 2N4957, JAN, JANTX, JANTXV, and JANS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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